

Supplement to: Classical (not Simulated!) Method of Moment Estimation of Life-Cycle Models

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Abstract

This pdf contains supplementary info to the paper. Mostly it details the model and estimation results for Gourinchas and Parker (2002). It also contains various details that make it easier to code GMM estimation for Life-Cycle Models.

Structural estimation of Life-Cycle models is often performed by the Simulated Method of Moments. We show how to instead use Generalized Method of Moments estimation, without the need for simulation. Simulation is here defined as the use of random number generators, in line with the theory underlying the Simulated Method of Moments. Since simulation introduces random noise, the Generalized Method of Moments estimator is asymptotically more statistically efficient than the Simulated Method of Moments estimator. For Life-Cycle models with one endogenous state the Generalized Method of Moments estimator is also computationally faster and has smaller numerical approximation error. We illustrate by reproducing the results of Gourinchas & Parker (2002), using Generalized Method of Moments estimation where they used Simulated Method of Moments estimation.

Keywords: Structural Estimation, Generalized Method of Moments, Life-Cycle model.

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This is a supplement to the paper that details the Gourinchas and Parker (2002) model and the estimation results. It also explains some things that help implement GMM in codes.

1 GMM with Seperable Moments

Let θ be the model parameters to be estimated and M_m be some model moments. Then we can consider the model as a function from θ to M_m , which we denote by $M_m(\theta)$.¹

We want to estimate the model parameters θ by targeting a vector of moments of the observed data, M_d . We need to assign a matrix of weights, W , that gives the relative importance of each of these moments. In practice, often only a subset of the model parameters are estimated, the rest might be calibrated in a step before estimation.²

Method of moments estimation aims to minimize the difference between the data moments and model moments by solving

$$\theta^* = \arg \min_{\theta} [M_d - M_m(\theta)]' W [M_d - M_m(\theta)] \quad (1)$$

Simulated method of moments replaces the model moments M_m , with simulated model moments $M_{m,s}$, and involves solving

$$\theta^* = \arg \min_{\theta} [M_d - M_{m,s}(\theta)]' W [M_d - M_{m,s}(\theta)]$$

Readers familiar with GMM will notice that this is not a standard formulation. Our definition only applies when the moment conditions are 'separable', which is the case in essentially all applications of GMM estimation to Life-Cycle models. The next paragraph provides the standard definition of GMM and shows how when the moment conditions are separable we get the above formulation.

Equation (1) is GMM, but does not make explicit the requirement that M_d is a moment, as well as only applying when the moment conditions are separable. For GMM, our moment conditions have to be population averages that evaluate to zero (which are implemented using the sample averages). Our formal moments conditions are $M_d - M_m = 0$, and not about M_d nor M_m separately. Method of moments requires the moment condition to be defined as $\frac{1}{N} \sum_{i=1}^N f(x_i, \theta) = 0$, that is as an average of a function of individual observations being equal to zero, where $\{x_i\}_{i=1}^N$ are the observed data, and the function $f(x_i, \theta)$ is defined as a (vector-valued) function of a single observation. Since M_d is a data moment, there must be a function $m()$ such that $M_d = \frac{1}{N} \sum_{i=1}^N m(x_i)$. So we can rewrite $M_d - M_m(\theta) = \frac{1}{N} \sum_{i=1}^N m(x_i) - M_m(\theta) = \frac{1}{N} \sum_{i=1}^N (m(x_i) - M_m(\theta))$, and since $(m(x_i) - M_m(\theta))$ is a

¹A formal proof providing some conditions under which a standard Life-Cycle model is a function from model parameters θ to model moments can be found, e.g., as part of Kirkby (2019).

² θ may include parameters that determine the exogenous shock processes, or the initial agent distribution, and may include the discount factor. Our notation omits these (potential) dependencies to make it easier to read.

function of a single observation x_i , we have that our formal moment condition is $\frac{1}{N} \sum_{i=1}^N (m(x_i) - M_m(\theta)) = 0$. Defining $g(\{x_i\}_{i=1}^N, \theta) = \frac{1}{N} \sum_{i=1}^N (m(x_i) - M_m(\theta))$, we can write the more formally precise definition of our GMM estimator as $\theta^* = \arg \min_{\theta} g(\{x_i\}_{i=1}^N, \theta)' W g(\{x_i\}_{i=1}^N, \theta)$.³ We use equation (1) as it is closer to both how estimating Life-Cycle models is commonly described, and to how the estimator is coded.

Note that for most GMM estimators it is not possible to rewrite the moment conditions as two separate terms, one which depends on the data, here $M_d = \frac{1}{N} \sum_{i=1}^N m(x_i)$, and one which is a constant (given θ), here $M_m(\theta)$. Notice that the first term is a random variable and the second term is just a constant, which will be important later when discussing efficient GMM. That the moment conditions are separable is not necessarily true when performing GMM estimation a Life-Cycle model but in practice it almost always is, including in all of the five papers we cited as examples of SMM estimation of Life-Cycle models in the Introduction.⁴

Regardless of whether doing GMM or SMM we have the same pseudo-code for the main optimization problem, as described in Algorithm 1,

Algorithm 1 Simulated/Generalized Method of Moments Estimation: Optimization

- | | |
|---|-------------------------------------|
| 1: Define M_d and W . | ▷ Estimate them from observed data. |
| 2: Guess initial θ . | |
| 3: Use optimization to solve $\min_{\theta} [M_d - M_{model}(\theta)]' W [M_d - M_{model}(\theta)]$ | |
-

There are a variety of optimization algorithms available to implement this and we will not discuss them here.⁵ The difference between Simulated Method of Moments and Generalized Method of Moments is in how we compute $M_{model}(\theta)$.⁶

SMM uses the pseudo-code in Algorithm 2 to calculate M_{model} , where N is the number of observations used to calculate the data moments M_d , and sN is the number of observations used to calculate the simulated moments M_d , with s a positive integer.⁷ There is a random number generator used in this process for drawing e_j and z_j .⁸ Implementing SMM requires that the random number generator used to compute $M_{model}(\theta)$ uses the same draws

³The language of the formal 'moment conditions' being made up a moment of the data and a moment of the model is clunky but standard. We are unfortunately not aware of a nicer terminology for describing this issue.

⁴Separability is better understood as a property of the moments, rather than the model, as some examples in simple (not Life-Cycle) models at github.com/robertdkirkby/GMMSeparableMoments illustrate. Note that it is still possible to GMM estimate Life-Cycle models in which the moment conditions are not separable, but separability allow us to more intuitively write the description of the problem to be estimated, and to simplify both the code to implement the estimation, and the implementation of efficient GMM.

⁵VFI Toolkit implements a number of options. An alternative to using optimization methods is to transform the problem to a Laplace-type estimator, a setting where MCMC is possible (Chernozhukov and Hong, 2003).

⁶The notation $M_{model}(\theta)$ corresponds to either the $M_m(\theta)$ or the $M_{m,s}(\theta)$ from earlier, depending on whether we are doing GMM or SMM.

⁷In our application all the target moments are age-conditional moments, and this code corresponds to calculating each data moment from N observations, and each simulated moment from sN observations. If our target was, e.g., total assets of all households, then it would be calculated from NJ data observations and sNJ simulated observations.

⁸A random number generator will typically also be required to draw from the initial agent distribution.

Algorithm 2 Simulated Method of Moments Estimation: $M_{model}(\theta)$

```
1: Given  $\theta$ .
2: Solve Life-Cycle model to get policy  $g_j(a, z, e; \theta)$ .
3: for  $ii = 1 : sN$  do
4:   Simulate a  $J$  period life cycle
5:   Draw initial agent state  $(a_1, z_1, e_1)$  from initial agent distribution.
6:   Store  $x_{i,1} = [a_1, z_1, e_1]$ 
7:   for  $j = 2 : J$  do
8:     Draw (random)  $e_j$  from  $\Pi_e(\cdot)$ 
9:     Draw (random)  $z_j$  from  $\Pi_z(\cdot|z_{j-1})$ 
10:     $a_j = g_{j-1}(a_{j-1}, z_{j-1}, e_{j-1}; \theta)$ 
11:    Store  $x_{i,j} = [a_j, z_j, e_j]$ 
12:   end for
13:   Store  $x_i = [x_{i,1}, x_{i,2}, \dots, x_{i,J}]$ .
14: end for
15: Calculate moments  $M_{model}(\theta) = \frac{1}{sN} \sum_{i=1}^{sN} f(x_i)$ 
16: return  $M_{model}(\theta)$ 
```

for evaluating each different value of θ .

By contrast, GMM uses the pseudo-code in Algorithm 3 to calculate M_{model} ,

Algorithm 3 Generalized Method of Moments Estimation: $M_{model}(\theta)$

```
1: Given  $\theta$ .
2: Solve Life-Cycle model to get policy  $g_j(a, z, e; \theta)$ .
3: Define grid on endogenous state  $a\_grid$ 
4: Use quadrature to discretize shocks:  $[z\_grid, Q_z] = discretize(\Pi_z)$  and  $[e\_grid, Q_e] = discretize(\Pi_e)$ .
5: Set period one distribution,  $\mu_1$ , equal to the initial agent distribution on the  $(a, z, e)$  grids.
6: for  $j = 1 : J - 1$  do
7:   Define  $T_j = g_j(\cdot, \cdot, \cdot; \theta) \circ Q_z \circ Q_e$ .  $\triangleright T_j$  is a Markov transition matrix on  $(a, z, e)$ 
8:    $\mu_{j+1} = T_j' \mu_j$ 
9: end for
10: Store  $\mu = [\mu_1, \mu_2, \dots, \mu_J]$ .
11: Calculate moments  $M_{model}(\theta) = \sum_{i=1}^{\#\mu} f(a, z, e, j) d\mu(a, z, e, j)$ 
12: return  $M_{model}(\theta)$ 
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where $\#\mu$ is the number of grid points in μ (which follows from the number of points in the a , z and e grids across all ages). Notice that there are *no* random draws here! We use quadrature to implement numerical integration instead of making random draws.

A few comments. (i) Notice that solving the policy function involves evaluating numerical integrals (as we have the *expection* of next period value function), and we assume that this is done by numerical quadrature as is standard in practice. (ii) The theory of SMM requires the same sequence from a random number generator to evaluate the model moments across different

values of θ .⁹ (iii) When coding the step $\mu_{j+1} = T'_j \mu_j$ of the GMM estimator the Tan improvement should be used to increase speed (by breaking this into two separate substeps, Tan (2020)). (iv) Depending on the details of the application, things like the quadrature step in the GMM will be independent of θ and so can be precomputed once rather than being done for each θ . (v) Depending on the algorithm used to get the policy, the grids and quadrature might be completed prior to the calculation of the policy.

We can perform these estimators with any (positive definite) weighting matrix W . The most asymptotically efficient estimator sets $W = \Omega^{-1}$, that is to use the inverse of the covariance matrix of the data moments as a weighting matrix. In this case the asymptotic variance for Generalized Method Moments simplifies to $(J'\Omega^{-1}J)^{-1}$ and for Simulated Method of Moments to $(1+1/s)J'\Omega^{-1}J$. So again, Simulated Method of Moments remains less efficient, with an asymptotic variance $1+1/s$ times as large.¹⁰ Ω is unknown so actually implementing this estimator will involve using the empirical estimate $\hat{\Omega}$, in place of Ω , so $W = \hat{\Omega}^{-1}$.

It turns out that for the kinds of moment conditions used in the estimation of Life-Cycle models, efficient GMM is easy to implement as $\hat{\Omega}$ simplifies to being the covariance matrix of the data moments and so can be calculated directly from the data without using the model. The key is the observation, made earlier, that our moment conditions are separable as $\frac{1}{N} \sum_{i=1}^N (m(x_i) - M_m(\theta)) = \frac{1}{N} \sum_{i=1}^N m(x_i) - M_m(\theta) = M_d - M_m(\theta)$, where the first term depends on the observed data and the second term is a constant. Therefore, Ω , which is the covariance matrix of our estimation moments $M_d - M_m(\theta)$, simplifies to just the covariance matrix of M_d , as $M_m(\theta)$ is a deterministic constant— $Cov(Z + c) = Cov(Z)$ for any random variable Z and constant c . So we can estimate $\hat{\Omega}$ as the covariance-matrix of our data moments, without any role for our model $M_m(\theta)$, and use (the inverse of) this estimate for our weighting matrix for efficient GMM. This provides an important contrast to SMM, as efficient SMM uses the covariance matrix of the moment conditions $M_d - M_{m,s}(\theta)$, but now $M_{m,s}(\theta)$ is a random variable (as it is simulated), and so we cannot calculate this from the data.

Instead two-iteration efficient SMM must be used. The first-iteration uses any weighting matrix, such as the identity matrix, to get a consistent estimate of θ , which we will call θ_1 . The second-iteration involves calculating the covariance matrix of the moment conditions $M_d - M_{m,s}(\theta_1)$, so based on θ_1 , and then using the inverse of this as the weighting matrix in the second-iteration. The second-iteration is required in SMM but not GMM as we have to account for the variance of the simulation and the covariance between the simulations and the data. Efficient estimation is substantially different for GMM and SMM estimation of Life-Cycle models, as the separability of the moment conditions means efficient GMM is done in a single step, whereas efficient SMM

⁹Modern algorithms for stochastic gradient descent could perhaps allow for this assumption to be relaxed but at the cost of introducing further noise into the estimator.

¹⁰McFadden (1989), Pakes and Pollard (1989), and Lee and Ingram (1991) all provide asymptotic variance results with both a generic W , and with $W = \Omega^{-1}$. Duffie and Singleton (1993) only provide results for $W = \Omega^{-1}$.

requires two iterations.^{11,12} So when efficient estimation is of interest—three of the five examples of SMM estimation of Life-Cycle papers mentioned in our Introduction implement the efficient estimator—the runtime gains from GMM over SMM can be substantial as they avoid the need for the second iteration, as well as substantially simplifying the estimation of $\hat{\Omega}$.

$W = \hat{\Omega}^{-1}$ is asymptotically efficient but it can be problematic in small samples (just as weighted least squares is more asymptotically efficient than OLS but typically performs worse in small samples). Altonji and Segal (1995) explain why $W = \hat{\Omega}^{-1}$ can be problematic in small samples: because the weighting matrix $W = \hat{\Omega}^{-1}$ may be correlated with the estimated moments, $\hat{M}_d - M(\theta)$, and this causes the estimator to be biased. They also provide intuition for when this is likely to happen, namely when our target moments include both first and second moments (rather than, e.g., just first moments as in our application to the model of Gourinchas and Parker (2002)).¹³ When using $W = \hat{\Omega}^{-1}$ is likely problematic (e.g., when we target both first and second-order moments), efficient GMM should be avoided; a popular alternative is to use $W = \text{diag}(\hat{\Omega})^{-1}$ (a matrix with zeros on the off-diagonals, and with the diagonal being the elementwise-inverse of the diagonal of Ω), although it is not clear this avoids the bias.¹⁴

To evaluate the asymptotic variance, Σ , we need J , W and Ω . W , and J can be easily computed once we have estimated the parameter vector,¹⁵ that leaves Ω . So we need an empirical estimate $\hat{\Omega}$ for the covariance of the moment conditions even if we do not use it's inverse as a weighting matrix.

¹¹In most situations the moment conditions for GMM are not separable in this manner, so efficient GMM cannot be done in a single step and two-iteration efficient GMM is needed. Life-cycle models are unusual, but not unique, in typically using separable moment conditions for estimation. All five of the example papers on SMM estimation of Life-Cycle models cited in the Introduction used moment conditions that are separable. This is better understood as a property of the moment conditions, rather than of the model, as illustrated by the examples at github.com/robertdkirkby/GMMSeparableMoments. It would be possible to use non-separable moments in the estimation of a Life-Cycle model, but is exceedingly rare in practice.

¹²Separable moment conditions mean that we can estimate the covariance matrix of the moment conditions without a value of θ , as it is independent of θ . But notice that we **cannot** evaluate the moment conditions themselves without a value of θ .

¹³For example, consider a setting where we estimate two moments of data $\{x_i\}_{i=1}^N$, the first is the mean of x , and the second is a variance of x . Call θ the parameter vector to be estimated, and let $g_1(\theta)$ be the mean for x of the model, $g_2(\theta)$ be the variance for x of the model. So our two moments for GMM are calculated as $\frac{1}{N} \sum_{i=1}^N (x_i - g_1(\theta))$ and $\frac{1}{N} \sum_{i=1}^N ((x_i - \bar{x})^2 - g_2(\theta))$. Consider the variance-covariance matrix of the moments, the inverse of which we would like to use as the weighting moment in efficient GMM. The first entry of the variance-covariance matrix is the 'square' of our first moment, namely $\frac{1}{N} \sum_{i=1}^N (x_i - g_1(\theta))^2$. This is similar to our second moment, and the two will typically be correlated in small samples. So we have that one of our moments, is correlated with an element of our weighting matrix, which is the inverse of this variance-covariance matrix. This is likely to occur in any situation where we use the first and second moments as targets, since the variance-covariance matrix element for the variance of the first moment is likely to correlate with our second moment targets. The examples in Altonji and Segal (1995) are about estimating covariance structures using GMM.

¹⁴Altonji and Segal (1995) advocate using $W = \mathbb{I}$ in small samples, but this means the parameter estimate is scale-dependent (e.g., different parameter estimates depending on whether we measure a moment in dollars or thousands of dollars, as the weights do not reflect the units). Using a W which is diagonal and avoids scale dependence can be better. Note that $W = \text{diag}(\hat{\Omega})^{-1}$ is conceptually subject to the same bias from the correlation between data moments and weighting matrix as causes small sample issues with $W = \hat{\Omega}^{-1}$.

¹⁵Recall that J is the Jacobian matrix of derivatives of the model moments with respect to the parameter vector, evaluated at the estimated parameter vector. This can be computed by taking finite differences of the model moments for small-epsilon changes in the parameter vector, or using automatic differentiation.

Results for GMM and SMM with time-series data are similar except we use T instead of N , and Ω (the covariance matrix of the moment conditions) is more complicated due to the presence of autocorrelation, and must be estimated with the Newey-West estimator or similar.

2 More about Runtimes

The actual ratio of runtimes will depend on hardware. Panel data simulations will be easily parallelized across CPUs and so runtimes drop (inverse) linearly with the number of CPUs. We report runtimes from a desktop with hardware that is reasonably balanced between CPU and GPU, in the sense the neither costs notably more than the other (6 CPUs and a GPU with 8GB of GDDR memory). A definitive comparison of runtimes is not attempted since differences not only impact run times but also the accuracy of moments, and both runtimes and accuracy will differ by the type of moment (mean, median, etc.). Ocampo and Robinson (2023) find that iteration combines more accurate moments with lower runtimes, relative to simulation, for longitudinal moments. Our experience is that agent distribution iteration provides a better speed/accuracy trade-off across essentially all one endogenous state Life-Cycle models and all moments. It is true for the variety of Life-Cycle models in the ['Intro to Life-Cycle Models'](#).

Note also that in situations where efficient GMM is of interest, the runtime gains for GMM over SMM are substantially increased. For GMM estimation of Life-Cycle models we can simply estimate $\hat{\Omega}$ from the observable data, whereas for SMM estimation of Life-Cycle models this is not possible and we must do two-iteration efficient SMM. So when we want the efficient GMM estimator, we can avoid the second-iteration which SMM would require. The runtime gain from not requiring two iterations is substantial, but not as high as fifty-percent as the first-iteration parameter estimate provides a good initial guess for the second-iteration and so the second iteration is much faster than the first.

3 Application to Gourinchas & Parker (2002)

We now use GMM to estimate the Life-Cycle model of Gourinchas and Parker (2002). Our estimates serve as a proof-of-concept. We use the *EstimateLifeCycleModel_MethodOfMoments()* command from VFI Toolkit, and the whole estimation requires less than 10 additional lines of code beyond those to setup and solve the model. The model was developed by Carroll (1997) and then estimated by Gourinchas and Parker (2002), and has been reestimated in the contexts of both understanding the sensitivity of parameter estimates to the estimation moments (Andrews, Gentzkow, and Shapiro, 2017) and of understanding the sensitivity of parameter estimates to the calibrated parameters (Jorgensen, 2023).

Gourinchas and Parker (2002) uses SMM to estimate the $J = 40$ period Life-Cycle model,

$$\begin{aligned} V_j(a, z, e) &= \max_{a'} v_j \frac{c^{1-\rho}}{1-\rho} + \beta E[V_{j+1}(a', z', e')|z] \\ \text{s.t. } c + a' &= Ra + ze \\ a' &\geq 0 \end{aligned}$$

where a is assets, c is consumption, v_j is a utility-shifter that depends on age j . Earnings follows an exogenous stochastic process comprised of a Markov z and an i.i.d. e . The terminal period value function is given by $V_{J+1} = \kappa v_{J+1} \frac{(X_{J+1} + H_{J+1})^{1-\rho_J}}{1-\rho_J}$, where $X_{J+1} = Ra_{J+1} + z_{J+1}$ and $H_{J+1} = h z_{J+1}$; by assumption $z_{J+1} = z_J$.^{16,17}

The (non-stationary) Markov shocks follow a non-stationary unit-root with drift,

$$\begin{aligned} \log(z_j) &= g_j + \log(z_{j-1}) + n \\ n &\sim N(0, \sigma_{z,n}) \end{aligned}$$

The i.i.d shocks are assumed to follow

$$e \begin{cases} \text{with probability } p : & e = 0 \\ \text{with probability } 1 - p : & u = \log(e), u \sim N(0, \sigma_u) \end{cases}$$

Agents in period $j = 1$ are distributed over assets as a log-normal distribution $\log(a) \sim N(\omega_{26,mean}, \omega_{26,stddev}^2)$. The article of Gourinchas and Parker (2002) does not discuss how the initial values of z and e are set. The code of Jorgensen (2023) sets period 0 to the mean value of $\log(Y_1)$, and then does one draw of the permanent shock innovations to get the period 1 permanent shock distribution; and uses the standard transitory shock distribution. The agent distribution is set to the cross-product of the asset, permanent shock, and transitory shock distributions.

Some parameters are calibrated, specifically g_j , $\sigma_{z,n}$, σ_u , $\omega_{26,mean}$, and $\omega_{26,stddev}$. We set all the parameters to their values from Gourinchas and Parker (2002) and report them in the Appendix.

The parameters β , ρ , h , κ and ρ_J are estimated by Generalized Method of Moments (by SMM in Gourinchas and Parker (2002)).

The target data moments are the log of the age-conditional mean of consumption (so $J = 40$ targets). We perform two estimates, following Gourinchas and Parker (2002). First, we do a robust

¹⁶Gourinchas and Parker (2002) do not explicitly include $a' \geq 0$ in their setup. But they follow Carroll (1997) in combining an assumption that agents must die with non-negative assets (that $a_{J+1} \geq 0$) with an income process that has a positive probability of taking a zero value in each period. Together these two assumptions imply that $a' \geq 0$ in every period.

¹⁷We make one change from the model estimated by Gourinchas and Parker (2002), namely they impose that $\rho_J = \rho$, while we allow them to differ. We discuss why this was done below when describing our estimates.

Estimated Parameters			
Robust estimates ($W = \mathbb{I}$)			
	<i>Point Estimate</i>	<i>90% Conf. Int</i>	<i>Description</i>
β	0.961	[0.952, 0.969]	discount factor
ρ	0.294	[0.053, 0.535]	curvature of utility fn
h	10.939	[-13.615, 35.494]	role of permanent income in warm-glow
κ	3.029	[0.258, 5.801]	relative importance of warm-glow
ρ_J	0.306	[0.138, 0.473]	curvature of warm-glow utility fn
Efficient estimates ($W = \Omega^{-1}$)			
	<i>Point Estimate</i>	<i>90% Conf. Int</i>	<i>Description</i>
β	0.961	[0.952, 0.969]	discount factor
ρ	0.297	[0.058, 0.535]	curvature of utility fn
h	37.459	[-37.781, 112.699]	role of permanent income in warm-glow
κ	2.927	[0.111, 5.742]	relative importance of warm-glow
ρ_J	0.286	[0.135, 0.438]	curvature of warm-glow utility fn

Table 1: GMM estimates for Gourinchas & Parker (2002)

estimate using $W = I$, i.e., the weighting matrix is the identity matrix. Second, we do an efficient estimate using $W = \hat{\Omega}^{-1}$, i.e., the weighting matrix has the inverse of the empirical variance of each moment on the diagonal, and zeros on the off-diagonal.¹⁸ Note that we can implement the efficient GMM directly, whereas Gourinchas and Parker (2002) had to use two-iteration efficient SMM.

Our estimates are reported in Table 3. Note that our results differ slightly from those of Gourinchas and Parker (2002) who estimate $\beta = 0.960$ and $\rho = 0.514$, and implicitly $h = \gamma_0/\gamma_1 = 0.0015/0.071 = 0.02111$.¹⁹

Our estimates differ because while they lay out precisely the model described above (on their pages 51-54) they do not estimate this exact model. Instead they convert the model into consumption Euler equations and solve these to get the policy function, instead of solving by value function iteration. While the switch from value function to consumption Euler equations does not change the solution, they do make a change to period $J + 1$ as part of the switch. When setting up the value function they parametrize V_{J+1} by κ and h , as above (but assuming $\rho_J = \rho$). For the consumption Euler equations they instead parametrize c_{J+1} by γ_0 and γ_1 .²⁰ Their paper provides the form for V_{J+1} that we use here (with $\rho_J = \rho$), and then derives their form for c_{J+1} under the assumption that retirees face an infinite horizon value function problem. In the data this assumption will fail, and so their parametrization of c_{J+1} will lead to different estimates from

¹⁸We take the empirical moments and their variances from the GitHub repo of Jorgensen (2023), who in turn got them from the replication materials of Andrews, Gentzkow, and Shapiro (2017). Note that this is the inverse of the variance-covariance matrix of the data moments; the zeros on the off-diagonal occur because the underlying dataset was a repeated cross-section and they are all age-conditional moments.

¹⁹Andrews, Gentzkow, and Shapiro (2017) got $\beta = 0.9574$ and $\rho = 0.6526$.

²⁰See their equation (6) which states that $c_{J+1} = \gamma_0 + \gamma_1 x_{J+1}$ (both c_{J+1} and x_{J+1} are lower case as they renormalize the model prior to solving). By definition $\gamma_0 = h\gamma_1$, but the relation between κ and the gammas is more complex.

the parametrization of V_{J+1} . Because the parametrizations of V_{J+1} and C_{J+1} are not compatible with each other, their solution and that presented here are making different assumptions about retirement (retirement is unmodelled but implicit in the period $J + 1$ parameterization). If both approaches used a non-parametric formulation for period $J + 1$ we would expect them to give the same solution (Jorgensen and To, 2020). But as they differ in the parametrization of period $J + 1$ it is unsurprising we get slightly different results. When estimating using the V_{J+1} , we originally imposed that $\rho_J = \rho$, we got not dissimilar point estimates, but much larger standard deviations, and looking at J (the derivatives of the moments with respect to the estimated parameters) the derivatives with respect to ρ are negative, while those with respect to ρ_J are positive. Based on this we decided it was better to estimate separate values for ρ and ρ_J , and that this was more in line with the original estimates of Gourinchas and Parker (2002) in terms of the parameterization of the final period.

This example has served three purposes. First, it provides proof-of-concept implementation that GMM estimation of Life-Cycle models is feasible. Second, we saw in the previous section that the runtime of the steps in evaluating a parameter vector for GMM is faster than for SMM. Third, it demonstrates how the efficient GMM estimator is simpler and faster —because the moment conditions are separable— than would be the case for SMM, where two-iteration efficient SMM had to be used by Gourinchas and Parker (2002).

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A Model Details

We described the model of Gourinchas and Parker (2002) in Section 3. This appendix just reports the values of all the pre-calibrated parameters which are set following Gourinchas and Parker (2002), and brief details of the computation.

Codes implementing the estimation (and the run-time results) can be found at:
<https://github.com/robertdkirkby/GourinchasParker2002>

Table A contains the calibrated values of the parameters of the model of Gourinchas and Parker (2002).

The parameters not mentioned in the table (other than those which will be estimated) are v and g . GP2002 set v , the age-dependent preference shifters, by first creating v_a as a fifth-order polynomial of age, and we get the polynomial coefficients from the code of Jorgensen (2023), and then some transformations of this get us to v itself: $v_a = 0 + 0.13964975 * age - 0.0047742190 * age^2 + 8.5155210 * 10^{-5} * age^3 - 7.9110880 * 10^{-7} * age^4 + 2.9789550 * 10^{-9} * age^5$. Then $v_{b,j} = \exp(\Delta_j v_{a,j})$, and finally set $v_1 = 1$ and iterate $v_j = v_{b,j} v_{j-1}^{1/(1-\rho)}$, for $j = 2, \dots, J$. Note that $v_{b,j} = (v_{j+1}/v_j)^{1/\rho}$ is all that is needed when solving the model using Euler equations as was done by previous studies (we have to use v , rather than v_b to be able to perform the value function iteration).

GP2002 set g , the age-dependent deterministic growth in earnings, based on the log-change in earnings as a fifth-order polynomial of age; the $\log(\text{earnings}) = 6.8013936 + 0.3264338 * age - 0.0148947 * age^2 + 0.000363424 * age^3 - 4.411685 * 10^{-6} * age^4 + 2.056916 * 10^{-8} * age^5$, and $g = \Delta_j \log(\text{earnings}_j)$.

Our initial values for the estimated parameters are $\beta = 0.9574$, $\rho = 0.6526$, $h = 0.0211$, $\kappa = 3$ (the first three are the estimates of Andrews et al. (2017), with $h = \gamma_0/\gamma_1$). Our initial value for ρ_J was set equal to the initial value for ρ .

Parameter	Value	Description
R	1.0344	Asset Returns
$\sigma_{z,n}$	$\sqrt{0.0212}$	Std. dev. of innovations to z
p	0.00302	Probability of zero earnings
σ_u	$\sqrt{0.0440}$	Std. dev. of e , when non-zero

Table 2: Calibrated Parameters of Gourinchas & Parker (2002)

The model is computed using VFI Toolkit. So the value function iteration is done with pure discretization, sped up with divide-and-conquer. The iteration of the agent distribution is done using the Tan improvement, as emphasized in the body of the paper. The optimization routine to minimize the GMM objective function is CMA-ES (Covariance Matrix Adaptation-Evolutionary Strategy), which is chosen as it is highly robust to non-linearities and local minima, although does

have slower runtimes than possible alternative algorithms. We discretize the i.i.d shocks using the an extension of the Tanaka-Toda method to handle age-dependent markov processes with drift. Note that we do not renormalize the problem to eliminate the unit-root markov shocks from the state space, which would be the cleaner/better/smarter approach but this model is small enough that we can afford to simply take a brute-force approach. We use 12 points to discretize the i.i.d. shocks, and 75 points to discretize the markov shocks; both are done with $nSigmas = 2$ (so plus/minus two standard deviations as the max/min grid points).

We recomputed the Jacobian matrix that determines the covariance matrix of the estimated parameter vector using larger grids, to ensure that our grids were large enough that numerical inaccuracy in the derivatives was not an issue.

Figure 1 shows the life-cycle profiles (age-conditional means) of consumption and income, both in the data, and for our estimates (the robust and efficient estimates have identical income processes, so only one is shown here), as well as the estimates for consumption of Andrews, Gentzkow, and Shapiro (2017).²¹ We computed the GMM objective function based on the consumption moments reported by Andrews, Gentzkow, and Shapiro (2017) and it is smaller than that based on the consumption moments associated with our robust estimates, however it appears to be impossible to do as well as the AGS2017 estimates for any parametrization of the model we are estimating (because of the differences in the parametrization of the final period).

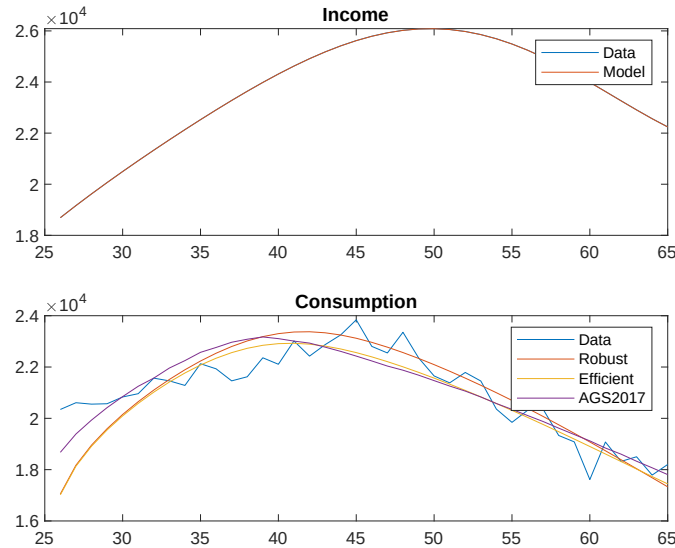


Figure 1: Age-conditional means of Income and Consumption in Data and Estimated Models

²¹ Andrews, Gentzkow, and Shapiro (2017) have similar point estimates for the estimated parameters to Gourinchas and Parker (2002), and the mean consumption profile of the former is available, while the later is not available to be best of our knowledge.